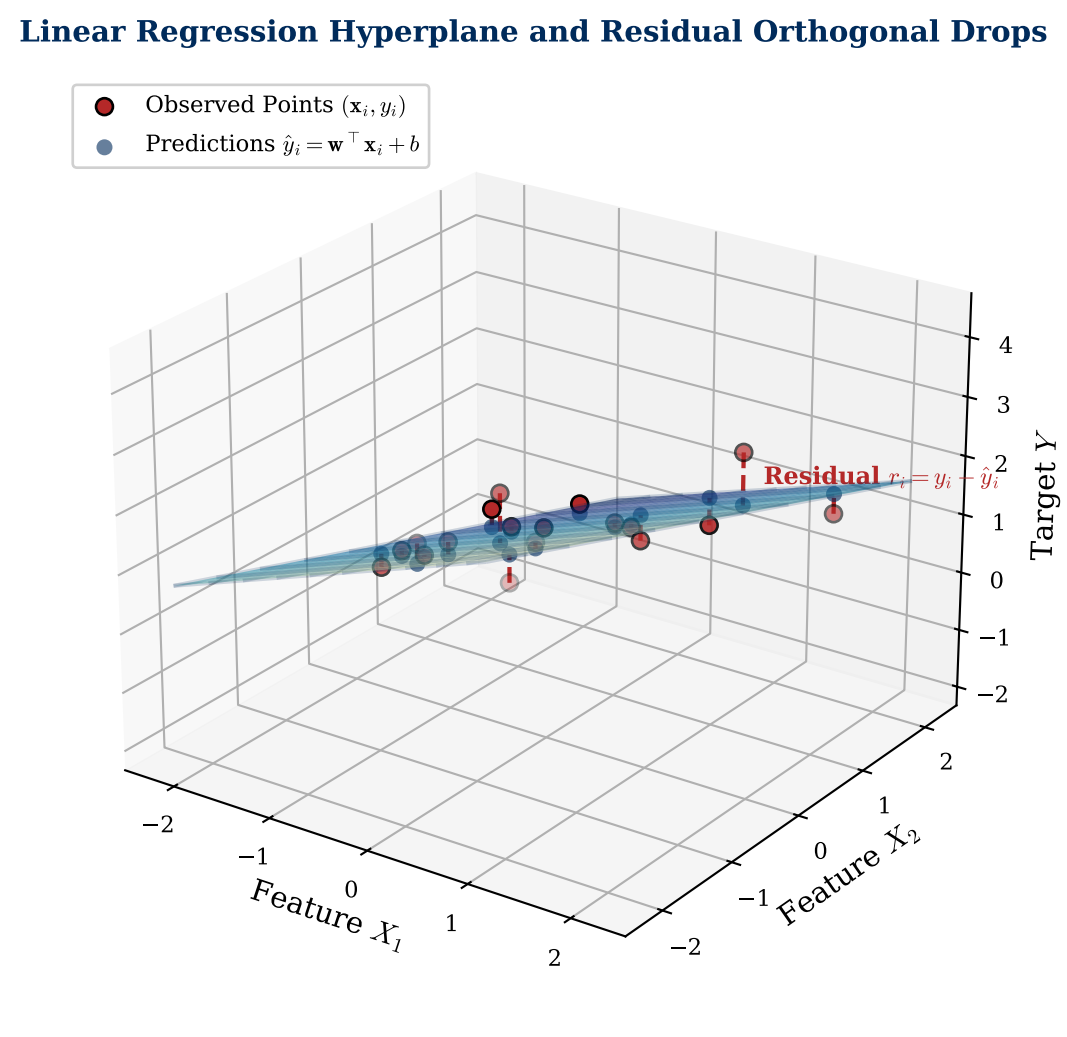

Linear Regression Hyperplane and Residual Orthogonal Drops
Observed Points $(\mathbf{x}_i, y_i)$
Predictions $\hat{y}_i = \mathbf{w}^\top \mathbf{x}_i + b$
Residual $r_i = y_i - \hat{y}_i$
Feature X_1
Feature X_2
Target Y
−2
−1
0
1
2
−2
−1
0
1
2
−2
−1
0
1
2
3
4